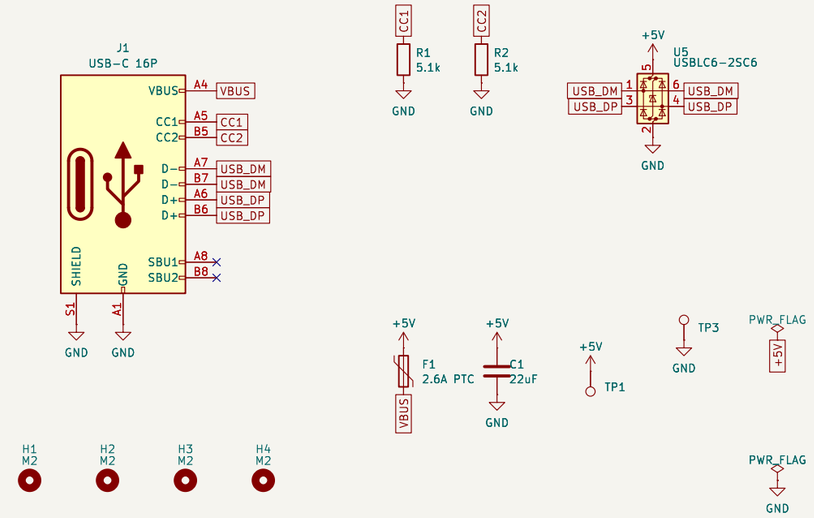
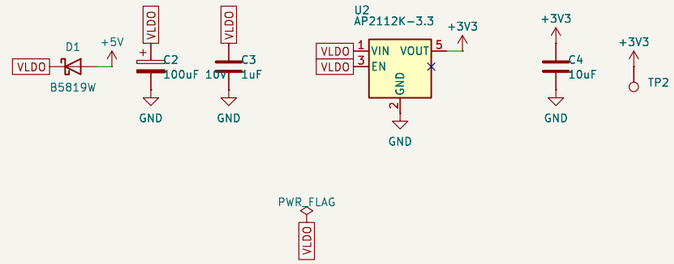
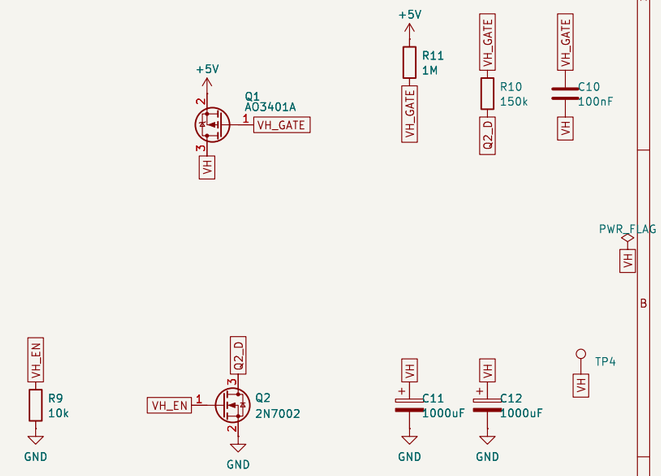
1 USB-C power input, 5 V



2 3.3 V logic supply

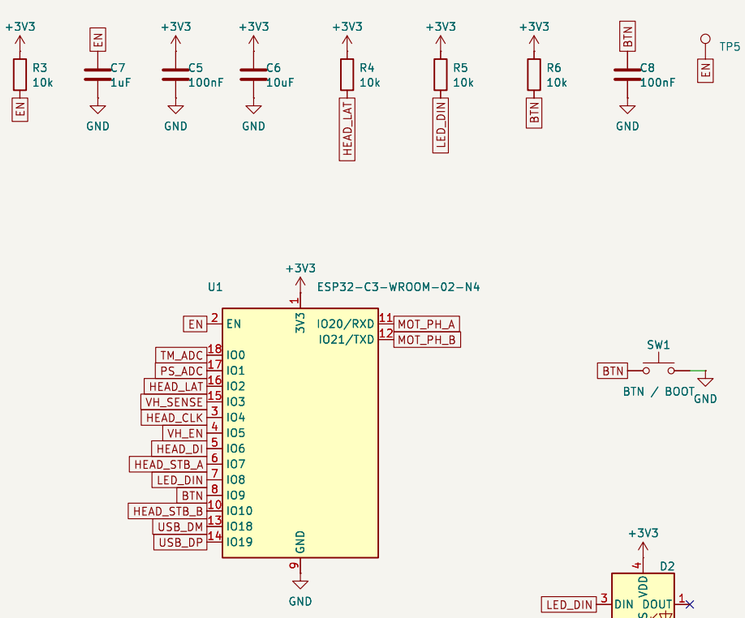


3 VH head-power switch with soft start

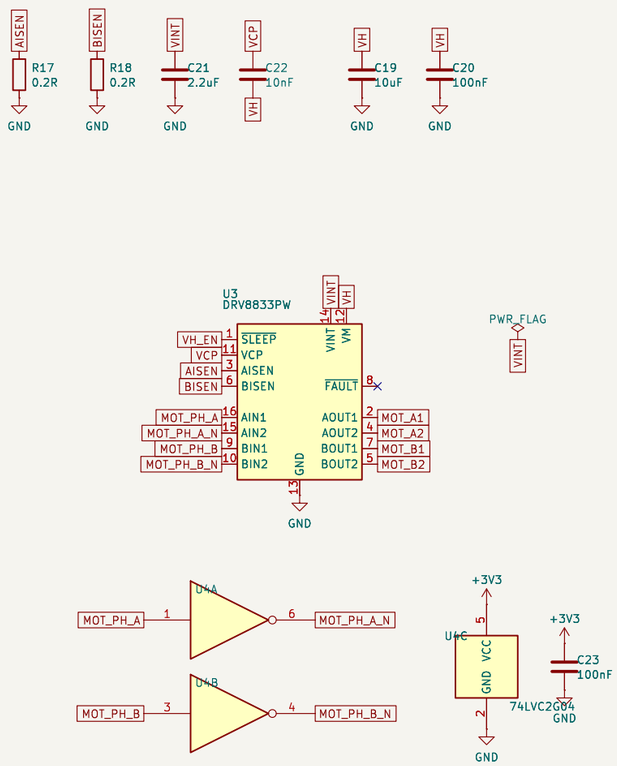


VH_EN high -> Q2 on -> Q1 on. C10/R10 slow the turn-on to ~20 ms to limit inrush into C11/C12.

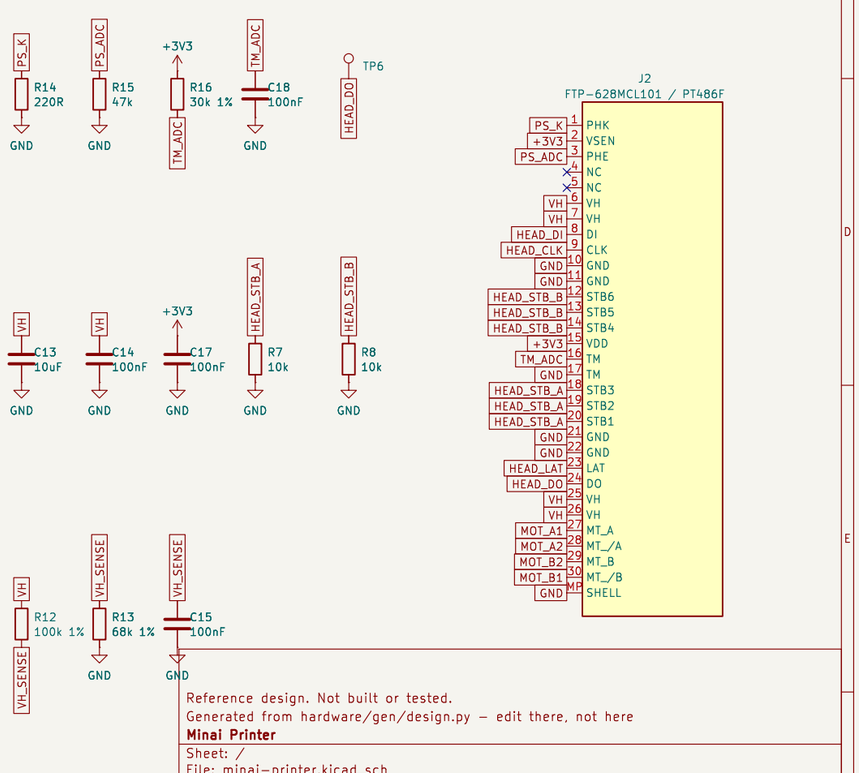
4 MCU: ESP32-C3 Wi-Fi / BLE module, button, status LED



5 Paper-feed stepper driver



6 Print mechanism connector, sensors, VH sense



REFERENCE DESIGN, NOT YET BUILT OR TESTED. J2 pinout follows the Fujitsu FTP-628MCL101 datasheet; confirm pinout, strobe polarity and FPC contact side against the mechanism actually purchased.

Strobes are assumed active-high (R7, R8 pull down). If the head is active-low, fit R7/R8 to +3V3 instead.

U4 makes IN2 = NOT IN1, so two MCU pins drive the motor in 2-phase full step. U3 sleeps when VH_EN is low.

Reference design. Not built or tested.
Generated from hardware/gen/design.py — edit there, not here
Minai Printer

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Title: *Minai Printer — main board reference design*

Size: A3	Date: 2026-11-10	**Rev: 0.1**
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